Dennis Delali Kwesi Wayo

Quantum Software Engineer — Quantum Systems Architect

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Summary

Research engineer building fault-tolerant quantum software for photonic and hybrid quantum architectures. Experience spans decoder benchmarking, architecture-level simulation, and reproducible scientific-computing pipelines in Python, Rust, and C++. Built end-to-end QEC tooling across hardware-to-decoder interfaces, with emphasis on decoder comparability, reproducibility, and architecture-level performance analysis. Current work sits at the boundary between scalable simulation, decoder infrastructure, and compiler-aware hardware modeling for fault-tolerant quantum systems.

Core Skills

Programming: Python, Rust, C++, Swift, Julia, Bash
Quantum Software: Calibration and benchmarking workflows, hardware-control interfaces, quantum error correction, surface-code threshold workflows, decoder benchmarking (BP/MWPM/UF), GKP digitization, photonic architecture modeling
Quantum Tooling: PennyLane, Qiskit, Qibo, Qibolab, Qibocal, Mitiq, QuTiP
Scientific Computing: Numerical linear algebra, HPC Linux workflows, parallel simulation, data pipelines, reproducibility and artifact tracking
Software Engineering: Git/GitHub, code review, CI/CD (GitHub Actions), testing (pytest), documentation, packaging, Docker, REST-style API development
ML for Science: PyTorch, TensorFlow, JAX, differentiable simulation

Quantum Software Engineering Experience

LiDMaS+ — Logical Decoder Benchmarking Engine (C++/Rust) 2026–Present

- Designed replay-driven QEC benchmarking pipelines with normalized decoder I/O to compare BP, MWPM, UF, and neural-style adapters on identical syndrome streams.
- Added diagnostics for syndrome satisfaction, residual nonzero rate, flip sparsity, and decoder stability, supporting reliable benchmarking and debugging across matched datasets.
- Implemented deterministic reproducibility checks (SHA-256 artifact matching) and backend APIs for provider onboarding, run tracking, and repeatable research-software delivery.

SchroSIM — Hardware-Agnostic Quantum Photonic Simulator (Swift/Rust) 2024–Present

- Built SchroSIM as a layered photonic-circuit platform separating circuit authoring, compilation, numerical execution, and hardware mapping for independent optimization and maintainable architecture.
- Implemented hybrid numerical pathways: efficient symplectic-state propagation for Gaussian blocks and higher-fidelity Fock-space/tensor-network style routines for non-Gaussian blocks.
- Developed Swift/Rust kernels and architecture-level interfaces for hardware-aware CV simulation, including GKP logical-state studies under loss and dephasing at the hardware/software boundary.

Research Assistant — Nazarbayev University 2021–Present

- Developed multiphysics and quantum-adjacent simulation workflows with finite-difference methods, HPC execution pipelines, and reproducible experiment structure.
- Built DFT/TDDFT simulation stacks (GPAW, ASE) for photonic material modeling and spectral analysis.

- Integrated ML-assisted scientific workflows with physics-based simulation, documented data handling, and reproducible experiment tracking.

Selected Open-Source Quantum Software Contributions

- **Qibolab** — Added a Dynamiqs simulation engine for emulator and hardware-control development workflows, with dependency integration and focused emulator tests; merged PR #1479.
- **Qibolab** — Fixed emulator evolution and result-handling paths after storage/GPU changes, including QuTiP/Dynamiqs compatibility and cyclic probability marginalization validated with Qibocal Rabi smoke runs; merged PRs #1497 and #1505.
- **Qibocal** — Fixed emulator example-platform drive configuration for current Qibolab schemas and validated Qibocal Rabi smoke runs on qubit/qutrit platforms; merged PR #1551.
- **Mitiq/Qibo** — Developed a Qibo tutorial combining repetition-code syndrome checks, post-selection/correction, noise sweeps, and zero-noise extrapolation for reproducible benchmarking workflows; open PR #3023.
- **Qiskit** — Extended `QuantumCircuit` standard and controlled-gate helpers to propagate gate labels, with regression tests #16373; refactored QPY roundtrip load-test setup #16372.
- **PennyLane** — Enabled documentation testing for `pennylane.shadows`, corrected executable classical-shadow examples, and added deterministic validation; merged PR #9566.
- **PennyLane** — Improved `qml.assert_equal` diagnostics for measurement-process mismatches with regression tests and changelog entry; open PR #9605.
- **PennyLane** — Added text-drawer multiplexer rendering for `SelectPauliRot` #9604 and improved `ExecutionConfig` print readability #9603.

Technical Leadership, Documentation & Community

- Develop maintainer-ready open-source contributions with focused tests, issue context, documentation updates, and PR narratives suitable for review by distributed quantum software teams.
- Prepare developer-facing examples and tutorials that translate quantum algorithms, error mitigation, calibration-adjacent workflows, and benchmarking concepts into executable software artifacts.
- Qiskit Global Summer School Development Week participant and scheduled QGSS mentor (2026); IBM Qiskit Advocate and QAMP Mentor (2025).

Education

PhD, Chemical Engineering 2026
Universiti Malaysia Pahang, FTKKP, Malaysia
Research focus: Quantum photonic-induced solid-state electrolyte for CO_2 reduction

M.Sc. Petroleum Engineering 2022
Nazarbayev University, Kazakhstan
B.Sc. Petroleum Engineering 2019
Kwame Nkrumah University of Science and Technology, Ghana
A.S. in Mechanical Engineering 2014
Tamale Technical University, Ghana

Current Graduate Study

PhD Candidate, Computer Science (extended research mode) Expected 2029
TU Bergakademie Freiberg, Faculty of Mathematics and Computer Science, Germany
Mode: Research-based and flexible; no taught-course classroom attendance requirement

Research focus: Parallel knowledge-graph and AI-augmented compiler architectures for scalable logical-GKP quantum error correction in continuous-variable quantum systems

M.S. Candidate, Computer Science (OMSCS) Expected 2027
Georgia Institute of Technology, College of Computing, Atlanta, USA
Specialization: Computing Systems; relevant preparation in algorithms, quantum computing/hardware, machine learning, NLP, HPC, high-performance computer architecture, software architecture & design, embedded systems optimization, and advanced operating systems

Selected Publications & Preprints

- Wayo, D.D.K. et al. (2026). A unified hardware-to-decoder architecture for hybrid continuous-variable and discrete-variable quantum error correction in LiDMaS+. *arXiv* 2604.15389.
- Wayo, D.D.K. & Groppe, S. (2026). RaCS: near-zero-error classical data encoding on photonic quantum processors. *Fortschritte der Physik*, 74(4), e70095.
- Wayo, D.D.K. et al. (2026). Decoder dependence in surface-code threshold estimation with native GKP digitization. *arXiv* 2603.25757.
- Wayo, D.D.K. et al. (2026). DifGa: Differentiable error mitigation for multi-mode gaussian and non-gaussian noise in quantum photonic circuits. *JPhys Photonics* accepted.
- Wayo, D.D.K. et al. (2026). Decoder Dependence in Surface-Code Threshold Estimation under Digitized Hybrid Continuous-Variable and Discrete Noise. *arXiv* 2603.06730. accepted
- Wayo, D.D.K. (2025). Simulation of ultrafast photonic circuits via nonlinear Schrödinger dynamics and quantum detector modeling. *Optical and Quantum Electronics*.
- Wayo, D.D.K. et al. (2025). Atomistic modeling of rare earth ions in photonic materials. *Luminescence*.
- Wayo, D.D.K. (2025). Ensembles of graph and physics-informed machine learning for scientific modeling. *Archives of Computational Methods in Engineering*.

Honors & Service

- Qiskit Global Summer School Development Week participant and scheduled QGSS mentor (2026)
- Womanium & WISER Quantum Scholar (2025); IBM Qiskit Advocate and QAMP Mentor (2025)
- Reviewer: Journal of Open Source Software, Scientific Reports, Nanoscale Advances